llmXive Automated Review of Latent Spatial Memory for Video World Models

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The paper makes several strong quantitative claims regarding efficiency and performance that require precise contextualization to be fully accurate.

First, the efficiency claims of **10.57x speedup** and **55x memory reduction** (Abstract, Introduction, Conclusion) are specific but potentially ambiguous regarding the baseline. The text correctly notes that the latent cache footprint shrinks by a factor of s^2 (where $s=16$, implying a 256x reduction in raw cache size). However, the reported **55x** reduction in *total* GPU memory suggests that other components (e.g., the diffusion backbone, VAE encoder/decoder buffers, or intermediate activations) constitute a significant portion of the memory budget in the baseline. The manuscript should explicitly define the "GPU memory" metric (e.g., "peak memory during the conditioning step" vs. "total rollout memory") and specify the exact baseline configuration (e.g., "Spatia with identical backbone") to ensure the 55x figure is not misleading. Without this, the claim risks overstating the benefit if the baseline comparison includes different model sizes or batch configurations.

Second, the reliance on **DepthAnything 3** (cited as ‘lin2025depth‘, ICLR 2026) and **Qwen3** (cited as ‘yang2025qwen3‘) for critical components (depth estimation and dynamic object filtering) presents a verification challenge. As this is a 2026 preprint, these citations refer to works that may not yet be publicly available or peer-reviewed in the standard sense. If these are hypothetical or "future-dated" citations used to justify the method's feasibility, the claims regarding the robustness of the depth estimation (Section 4.2, Table 4) are currently unverifiable by the community. The authors should clarify the status of these references or provide a link to the specific preprint versions used.

Finally, the claim of **state-of-the-art (SOTA) performance** on WorldScore (Abstract) is technically supported by Table 1, where Mirage (70.36) exceeds Spatia (69.73). However, the margin is narrow (~0.6 points). While "SOTA" is factually correct if no other method scores higher, the phrasing in the abstract ("attains state-of-the-art performance") might benefit from a qualifier (e.g., "sets a new SOTA" or "achieves competitive SOTA") to reflect the modest improvement over the closest 3D-aware baseline, preventing readers from inferring a large performance gap.

- [writing] The claim of '55x lower GPU memory usage' (Abstract, Intro, Conclusion) lacks a clear baseline definition. Does this compare total system memory or only the cache footprint? The text states the cache shrinks by s^2 (256x), so a 55x total system reduction implies significant overhead elsewhere. Clarify the exact metric and baseline to support this specific magnitude.
- [science] Citations for 'DepthAnything 3' (lin2025depth) and 'Qwen3' (yang2025qwen3) refer to 2025/2026 publications. As a reviewer of a 2026 preprint, verify these are real, accessible works. If these are hypothetical or future-dated citations not yet public, the claims relying on them (e.g., depth estimation quality) are currently unverifiable.
- [writing] The claim that Mirage achieves 'state-of-the-art performance on WorldScore' (Abstract) is supported by Table 1, but the table shows Mirage (70.36) is only marginally higher than Spatia (69.73). Ensure the term 'state-of-the-art' is qualified (e.g., 'competitive' or 'new SOTA by X points') to avoid overstating the margin of improvement.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
The figure effectively illustrates the architectural difference between the proposed method and prior systems. However, the top diagram uses the generic label 'Spatial Memory' for the RGB point cloud stage, which contradicts the specific terminology used in the caption.
Figure 2
The figure provides a clear visual pipeline of the initialization and chunk-based generation process. However, the caption contains significant grammatical errors, including a missing subject in the first sentence and a truncated final sentence.
Figure 3
The figure effectively visualizes the comparison between methods, but the caption is poorly written with missing subject names (e.g., 'generalizes beyond...') that make it difficult to map specific claims to the labeled rows.
Figure 4
The figure effectively visualizes the efficiency gap between methods, but the caption contains a truncated sentence and a potentially confusing claim regarding the memory growth rate of the baselines versus the proposed method.
Figure 5

The figure effectively visualizes the comparison across methods and time, but the caption's claim of showing 'one trajectory' per block is confusing given the three distinct scenes shown without clear delimiters.

Figure 6

The figure displays a standard comparison grid but fails to visually demonstrate the specific 'closed-loop revisit' claim made in the caption, as it does not show the final frame compared to the input frame. Additionally, the caption contains a missing subject in the final sentence.

Figure 7

The figure effectively displays a visual comparison of video generation methods on an indoor trajectory, but the caption is grammatically incomplete, failing to name the specific methods being compared in the text.

- [writing] Figure 1: The top diagram labels the intermediate 3D representation as 'Spatial Memory', but the caption explicitly defines this as 'RGB point cloud based memory'. The diagram should use the specific term 'RGB Point Cloud' to match the caption's distinction.
- [writing] Figure 2 caption: The phrase 'Overview of .' contains a missing subject (likely the method name) immediately following the preposition, rendering the sentence grammatically incomplete.
- [writing] Figure 2 caption: The final sentence ends abruptly with 're-rasterises the accumulated', missing the object noun (e.g., 'memory' or 'cache') and the closing period.
- [writing] Figure 3: The caption contains multiple grammatical errors where the model name is missing (e.g., 'generalizes beyond...', 'RGB point cloud baselines show...'). The text should explicitly name the proposed method (e.g., 'Ours' or the paper title) to match the figure labels.
- [writing] Figure 3: The caption claims 'foundation video generators drift in geometry' but does not explicitly identify which row corresponds to this baseline (likely Voyager), making the specific claim hard to verify against the visual evidence.
- [science] Figure 4: The caption claims 'peak cache footprint... grows by less than 0.5 MiB per chunk' for the proposed method, but the right chart shows the 'Gen3C' baseline (light teal) growing from 22.3 to 43.8 MiB (a ~21 MiB jump) and the 'Spatia' baseline (medium blue) growing from 23.4 to 47.0 MiB (a ~23.6 MiB jump). The text likely intended to describe the 'Ours' method (red bar), but the phrasing is ambiguous and the data for the baselines contradicts the 'less than 0.5 MiB' claim if applied genera
- [writing] Figure 4: The caption text is truncated at the end ('...re-rasterises the accumulated'), cutting off the sentence before the period or the figure file reference.
- [writing] Figure 5: The caption states 'Each block shows one RealEstate10K trajectory', but the image displays three distinct scenes (indoor, outdoor house, outdoor pool) without clear visual separators or labels to distinguish these separate trajectories.

- [writing] Figure 5: The caption claims the method 'preserves sharper structure' but does not explicitly name the method (e.g., 'Latent Spatial Memory' or 'Ours') in the sentence, relying on the reader to infer the subject from the row label.
- [science] Figure 6: The caption claims a comparison between the 'last frame' and the 'input frame' to demonstrate consistency, but the figure displays a sequence of 5 frames (Input Frame, Spatia, GEN3C, Ours) without explicitly labeling which is the 'last frame' or showing the revisit result side-by-side with the input.
- [writing] Figure 6: The caption contains a grammatical error where the subject is missing in the phrase 'shows that maintains strong consistency'; it should specify the method (e.g., 'shows that Ours maintains...').
- [writing] Figure 7 caption: The sentence 'maintains coherent layout...' lacks a subject; it should explicitly name the proposed method (e.g., 'Ours' or the paper title) to match the visual rows.
- [writing] Figure 7 caption: The sentence 'whereas baselines suffer from...' lacks a subject; it should explicitly name the baseline methods (e.g., Voyager, Spatia, VMem) to clarify which rows correspond to the described failures.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating barriers for non-specialist readers. In the Abstract, terms like "VAE" (Variational Autoencoder) and "RGB" are used immediately without expansion. While "VAE" is common in generative modeling, its unexplained use in the opening sentence assumes a level of specialization that the paper's broad claims about "video world models" might not warrant. Similarly, "RGB" should be spelled out or contextualized as "pixel-space color" to contrast effectively with the proposed "latent" approach for a general audience.

In Section 1 (Introduction), the term "ControlNet-style side branch" is introduced without defining what ControlNet is. A reader unfamiliar with this specific architecture for conditional generation will not understand the mechanism being described. In Section 3.4, "LoRA" (Low-Rank Adaptation) is used without definition. While standard in LLM fine-tuning, its application here to video diffusion backbones is a specific technical choice that requires a brief expansion for clarity.

Furthermore, "WorldScore" is cited as a benchmark in the Abstract and Introduction without defining what it measures or why it is the standard. While the citation provides the source, the text itself treats the metric as a known quantity, which may confuse readers from other fields. The term "flow-matching" in Section 3.4 and the Appendix is also used without a brief parenthetical explanation of the objective function, assuming the reader knows the specific optimization landscape.

Finally, the phrase "rasterize-and-encode round trip" in the Abstract and Section 1 is a dense technical description. While accurate, it could be slightly softened to ”rendering and re-encoding cycle” to improve flow, though the primary issue remains the unexplained acronyms. The paper would benefit from a ”Glossary of Terms” or simply ensuring every acronym (VAE, RGB, LoRA, ControlNet, WorldScore, VACE, SAM3, Qwen3) is fully spelled out upon its first appearance in the main text.

- [writing] Define ’VAE’ at first use in the Abstract and Introduction. While standard in the field, the term is used immediately without expansion, which excludes readers from adjacent disciplines.
- [writing] Replace the acronym ’RGB’ with ’red-green-blue color’ or ’pixel color’ on first occurrence in the Abstract and Section 1 to ensure clarity for non-specialists.
- [writing] Define ’ControlNet’ upon first mention in Section 1. The text assumes the reader knows this specific architectural pattern for injecting conditions.
- [writing] Replace the acronym ’LoRA’ with ’Low-Rank Adaptation’ at its first appearance in Section 3.4. The current usage assumes prior knowledge of this specific fine-tuning technique.
- [writing] Define ’WorldScore’ at first mention in the Abstract. The text treats it as a known entity, but it is a specific benchmark introduced in the cited work, not a universal standard.

paper_reviewer_logical_consistency

Verdict: minor_revision
The logical consistency of the paper is generally strong regarding the core mechanism: storing latent features avoids the lossy and expensive round-trip of rendering RGB and re-encoding. The derivation of the readout operation (Eq. 3) logically follows from the initialization (Eq. 2), and the ablation studies (Table 4) provide coherent evidence that the latent representation is superior to the RGB alternative for the specific backbone used.

However, there is a significant logical gap in the explanation of the **memory footprint reduction**. The text repeatedly claims the cache footprint shrinks by the ”squared VAE compression factor” (e.g., $16^2 = 256$). This is mathematically inconsistent with the definitions provided. The RGB point cloud stores 3 float values (RGB) per point, while the latent memory stores C float values (e.g., $C=48$ in Sec 4.1). Therefore, the latent cache is actually **larger** per point by a factor of $C/3$ (approx. 16x), not smaller. The massive memory savings (55x) likely arise from the elimination of the **high-resolution rasterization buffer** and the **VAE encoder’s intermediate feature maps** during the conditioning step, not from the storage size of the cache points themselves. The current phrasing conflates ”cache storage size” with ”total GPU memory usage during the readout step,” which misrepresents the source of the efficiency gain.

Additionally, the causal link for the **10.57x speedup** requires clarification. The method performs a ”decode-and-re-encode” step to update the cache (Algorithm 1, lines 13-

16). The baseline performs a ”rasterize-and-encode” step. The paper asserts the baseline is slower but does not explicitly demonstrate that the *update* cost in the proposed method is negligible compared to the *readout* cost in the baseline. If the update step is frequent and expensive, the net speedup might be lower than claimed unless the baseline's rasterization overhead is proven to be orders of magnitude higher than the proposed method's decoding overhead. The current argument assumes the baseline's bottleneck is the readout loop without fully accounting for the proposed method's own update loop in the comparative analysis.

- [science] The claim that cache footprint shrinks by 'squared VAE compression factor' is logically inconsistent. Latent features (C=48) are larger per point than RGB (3 channels). The 55x saving likely comes from avoiding high-res rasterization buffers and encoder activations during readout, not cache storage size. Clarify this distinction.
- [science] The 10.57x speedup claim relies on the baseline's 'rasterize-and-encode' being dominant. However, the proposed method also performs 'decode-and-re-encode' for updates. The paper does not explicitly isolate the cost of the update step in the baseline comparison, leaving a gap in the causal chain for the specific speedup magnitude.

paper_reviewer_overreach

Verdict: minor_revision
The paper presents a compelling architectural shift by moving spatial memory from RGB space to the latent space of the diffusion model. However, several quantitative claims regarding efficiency and performance appear to overreach the provided evidence or lack necessary context.

First, the headline efficiency metrics of ”10.57x faster end-to-end video generation” and ”55x lower GPU memory usage” (Abstract, Introduction) are potentially misleading. The efficiency analysis in Section 4.3 and Figure 4 isolates the ”cache read” time and ”cache footprint.” While the readout cost is indeed significantly lower, the ”end-to-end” generation time is dominated by the diffusion denoising steps, which are computationally identical for both the proposed method and the RGB-baseline (assuming the same backbone and number of steps). A speedup in the readout component (which is a fraction of the total time) cannot mathematically result in a 10x speedup of the total generation time unless the number of denoising steps is also reduced or the chunking strategy fundamentally changes the total compute. The authors must clarify if ”end-to-end” refers to the total wall-clock time or just the memory-conditioning loop, and provide a full timing breakdown to support the 10.57x claim.

Similarly, the ”55x” memory reduction claim is ambiguous. It is unclear if this refers strictly to the memory footprint of the 3D cache structure (which is expected to be smaller due to latent compression) or the total peak GPU memory usage during generation. If the latter, the comparison is likely unfair if the baseline's total memory includes the VAE encoder and rasterization buffers that are bypassed in the proposed method's readout, but the proposed method still requires the full backbone and VAE decoder for generation. The authors should explicitly define the scope of the memory measurement (e.g., ”peak memory excluding backbone weights” or ”cache-only memory”) to ensure a fair comparison.

Finally, the claim that the method ”avoids the per-step pixel-space detour” (Section 3.1) is slightly overstated. While the *readout* phase successfully avoids this, the *cache update* phase (Section 3.4, Algorithm 1) explicitly requires decoding generated frames to RGB, estimating depth, and re-encoding them to latents to update the memory. This pixel-space round-trip still occurs, albeit amortized over a chunk. The text should be refined to state that the *conditioning loop* is free of this cost, rather than implying the entire pipeline avoids it, to maintain scientific precision.

- [writing] The claim of ’55x lower GPU memory usage’ is ambiguous. Clarify if this refers to the cache structure alone or total peak memory (including backbone/VAE). If total, the comparison may be unfair as the baseline includes VAE encoder overhead in the readout loop.
- [science] The ’10.57x faster end-to-end’ claim likely conflates readout speed with total generation time. Since denoising steps dominate total time and are identical for both methods, a 10x readout speedup cannot yield a 10x total speedup. Provide a full end-to-end timing breakdown.
- [writing] The claim that the method ’avoids the per-step pixel-space detour’ is overstated. The cache update step (Sec 3.4, Alg 1) still requires decoding to RGB and re-encoding. Clarify that only the conditioning loop avoids this cost, not the entire pipeline.

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents a technical advancement in video world models by shifting spatial memory from RGB space to latent space. From a safety and ethics perspective, the primary concerns revolve around the reliance on third-party foundation models for critical geometric and semantic filtering, and the implications of generating open-domain content.

First, the method depends heavily on external models for dynamic object filtering and depth estimation. Specifically, Section 3.4 and Algorithm 1 cite ”Qwen3-VL” and ”SAM3” to segment dynamic objects and sky regions before updating the persistent memory. The paper does not disclose the training data or potential biases of these specific foundation models. If these models exhibit bias against certain visual patterns, demographics, or cultural contexts, the ”dynamic object filter” could systematically erase or misrepresent these elements in the generated world, leading to a form of algorithmic erasure. The authors should explicitly discuss the provenance of these auxiliary models and any known limitations regarding bias or fairness in their segmentation capabilities.

Second, the paper claims the model generalizes to ”open-domain” prompts (Figure 5) despite being trained on the RealEstate10K dataset. While the results show geometric consistency, the safety implications of generating arbitrary open-domain video content are not addressed. Without explicit content moderation or safety filters, such a system could be used to generate non-consensual deepfakes, copyrighted material, or harmful scenarios. The authors should clarify if any safety guardrails or content filters are applied during inference to prevent the generation of prohibited content.

Finally, the reliance on ”feed-forward reconstructors” (DepthAnything 3) for metric depth introduces a potential point of failure for safety-critical applications. If the depth estimation is inaccurate, the resulting 3D memory could be geometrically flawed, potentially leading to hazardous outcomes if this technology were deployed in autonomous systems. The paper should briefly discuss the robustness of the depth estimation component and whether uncertainty quantification is considered to mitigate risks in real-world deployment.

Overall, while the technical contribution is sound, the manuscript requires a brief discussion on the ethical implications of its reliance on biased foundation models and the safety measures required for open-domain generation.

- [writing] The manuscript relies on 'open-vocabulary entity extractors' (Qwen3-VL) and 'video segmenters' (SAM3) to filter dynamic objects (Sec 3.4, Alg 1). Explicitly state the training data sources and potential biases of these foundation models, as they may systematically misclassify specific demographics or cultural contexts as 'dynamic' or 'sky,' leading to erasure in the generated world model.
- [writing] The paper claims to generate 'open-domain' videos (Fig 5) using a model trained on RealEstate10K. Clarify the safety protocols or content filters in place to prevent the generation of harmful, non-consensual, or copyrighted content when the model is prompted with open-ended queries outside the training distribution.
- [writing] The method uses a 'feed-forward reconstructor' (DepthAnything 3) to estimate metric depth for memory construction. Discuss the potential for safety-critical failures if the depth estimation is inaccurate in real-world deployment scenarios (e.g., autonomous navigation), and whether the system includes uncertainty quantification to mitigate such risks.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence supporting the core claims of latent spatial memory is generally robust, particularly regarding the architectural efficiency gains. The ablation studies in Section 4.4 (Table 3) provide strong causal evidence that the latent representation and dynamic filtering are necessary for the observed performance, as removing them causes significant drops in consistency scores (e.g., Dynamic Score dropping from 67.11 to 59.70). The efficiency claims in Section 4.3 are well-supported by the theoretical analysis in the Appendix (Section 2) regarding the removal of the VAE encoding step from the critical path.

However, the statistical rigor of the main benchmark results requires clarification. In

Table 1 (WorldScore), the improvement over the strongest baseline (Spatia) in the primary 'Average Score' metric is marginal (70.36 vs 69.73). While the ablation suggests the method is sound, the paper does not report variance, standard deviation, or p-values for these benchmark scores. Given the stochastic nature of diffusion models, it is unclear if the 0.63 point gain is statistically significant or within the noise floor of the evaluation protocol. Similarly, the RealEstate10K results in Table 2 present a mixed signal: while PSNR_C improves, the LPIPS_C (a perceptual metric often more aligned with human judgment) is slightly worse (0.228 vs 0.213). The text glosses over this by focusing on PSNR, which may not fully capture the perceptual consistency trade-offs.

Furthermore, the efficiency metrics (10.57x speedup, 55x memory reduction) are presented as single-point measurements on a single H100. While the theoretical scaling is sound, the lack of multiple runs or error bars makes it difficult to assess the stability of these gains under different system loads or rollout lengths. The authors should provide statistical context (mean ± std) for the efficiency measurements and clarify the significance of the marginal WorldScore improvements.

- [science] Report standard deviation or confidence intervals for the efficiency claims (10.57x speedup, 55x memory) in Section 4.3. Clarify if the single H100 measurement is a mean of multiple trials or a single run.
- [science] Clarify the discrepancy between the small WorldScore gain (0.63 pts) and the large ablation drop (7.4 pts) for the dynamic filter. Provide statistical significance tests (e.g., t-test) to confirm the main result is not within noise.
- [science] Address the trade-off in Table 2 where LPIPS_C degrades (0.228 vs 0.213) despite PSNR_C improvement. Explain why the PSNR gain outweighs the perceptual metric loss or provide statistical justification.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical rigor of the experimental evaluation in Section 4 requires clarification to support the quantitative claims made. While the methodological design is sound, the reporting of results lacks necessary statistical context.

First, the efficiency claims in Section 4.2 ("Efficiency") state that the method is "10.57x faster" and uses "55x less GPU memory." These figures are presented as precise constants derived from measurements on a single NVIDIA H100. However, no standard deviation, confidence intervals, or number of trials are reported. In computational experiments, wall-clock time can vary significantly due to system noise, memory fragmentation, or background processes. Without reporting the variance (e.g., mean ± std dev over $N \ge 5$ runs), it is impossible to determine if the observed speedup is statistically significant or if the "order of magnitude" claim holds under rigorous testing. The authors should re-run the efficiency benchmarks with multiple seeds and report the distribution of results.

Second, the main performance metrics in Table 1 (WorldScore) and Table 2

(RealEstate10K) are presented as single scalar values. The paper does not specify the number of independent generation seeds used to compute these averages. Video generation models often exhibit high variance in output quality depending on the random noise initialization. A difference of 0.63 points in the WorldScore Average Score (70.36 vs 69.73) may not be statistically significant without knowing the standard error. The authors must explicitly state the number of seeds used (e.g., "results averaged over 5 seeds") and ideally provide error bars or confidence intervals in the tables or text.

Finally, the ablation studies in Table 3 isolate components by disabling them. The results show performance drops (e.g., "No Dynamic Object Filter" drops the score to 61.20). However, these comparisons are also presented as single-point estimates. To validate that these components are genuinely necessary and not just benefiting from a lucky random seed in the full model, the authors should report the variance across seeds for the ablated variants as well. If the full model and the ablated model have overlapping confidence intervals, the claim of "degradation" is not statistically supported.

The paper currently relies on point estimates that may be subject to high variance. Rerunning experiments with multiple seeds and reporting statistical dispersion is required to substantiate the claims of state-of-the-art performance and efficiency.

- [science] Report standard deviation or confidence intervals for the efficiency metrics (10.57x speedup, 55x memory reduction) in Section 4.2. Single-point measurements on a single H100 without variance estimates make the statistical significance of the performance gains unverifiable.
- [science] Clarify the statistical protocol for the WorldScore and RealEstate10K results. Specify the number of independent seeds used for generation and whether the reported scores are means over multiple runs. Without this, the observed margins (e.g., 70.36 vs 69.73) cannot be assessed for significance.
- [science] In the ablation study (Table 3), provide error bars or p-values for the performance drops. The claim that 'No Dynamic Object Filter' degrades stability relies on a single run; statistical validation is needed to confirm the effect is not due to random generation variance.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript is generally well-written, with a clear logical flow and precise technical vocabulary appropriate for the field. The abstract and introduction effectively set up the problem and the proposed solution. However, there are several minor grammatical errors and instances of slightly awkward phrasing that detract from the overall polish.

Specifically, in Section 4.2 ("Main Results"), there is a subject-verb agreement error where "Figures... provides" should be "Figures... provide". Additionally, the sentence "foundation video generators without spatial memory drift noticeably" is syntactically ambiguous; it would be clearer to state that these generators "exhibit noticeable drift" or simply "drift noticeably." The phrase "The gap widens with horizon" also lacks a necessary article, which should be corrected to "with the horizon" or "with increasing horizon."

In Section 4.3 ("Ablation Studies"), the sentence "This confirms that bottleneck of the RGB detour" is missing the definite article "the" before "bottleneck." While these issues do not fundamentally obscure the scientific meaning, correcting them will improve the readability and professional quality of the paper. The rest of the text is coherent and free of major grammatical faults.

- [writing] In Section 4.2 (Main Results), the sentence 'Figures~\ref{fig:re10k_video} and~\ref{fig:re10k_revisit} provides...' contains a subject-verb agreement error. The plural subject 'Figures' requires the verb 'provide' instead of 'provides'.
- [writing] In Section 4.2, the phrase 'foundation video generators without spatial memory drift noticeably' is syntactically ambiguous. It is unclear if 'drift' is a noun or verb here. Rephrase to 'drift noticeably' or 'exhibit noticeable drift' for clarity.
- [writing] In Section 4.2, the sentence 'The gap widens with horizon' is missing an article. It should read 'The gap widens with the horizon' or 'with increasing horizon' to be grammatically complete.
- [writing] In Section 4.3 (Ablation Studies), the phrase 'This confirms that bottleneck of the RGB detour' is missing a definite article. It should read 'This confirms that the bottleneck of the RGB detour'.